# Braid Arrangements for Concurrent Programs

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**Abstract**

Precubical sets are a standard tool in the study of concurrent programs. A 2021 paper by Paliga and Ziemiański showed that the fundamental groupoid path space for the final precubical set was exactly the category Braid. However, they also showed induced braid subgroups for euclidean sets trivialized. In the world of hyperplane arrangements, this kind of trivialization can be resolved by working in the "complexified" space. We use this same approach, to build a new, complexified version of the path space, $\mathrm{Ch}^*$. This is homotopy equivalent to the Salvetti Complex of the braid arrangement in for the cube, but defined over any precubical set. By Salvetti's theorem, then, $\pi_1|\mathrm{Ch}^*(\square^n)|$ is is the pure braid group. This gives us a new braid invariant for concurrent programs.

## Contents



## 1 Introduction

This work is in 3 parts. First we'll recap the relevant material in brief. Second, we will perform the construction of the "complexified" path space. Third, we will study some of its properties.

To begin let's fix some notation, let Cube be the category of cubes, with face embeddings as morphisms. Let $\square$ be the precubical sets, defined as the presheaf topos over Cube. A $\square$ set then has, for each $n \in \mathbb{N}$, a set of $n$-cells denoted $K[n]$. And, for each $n \in \mathbb{N}$, a face map $d_i^\epsilon$, where $i$ indicates the direction, and $\epsilon$ is a boolean, with coherence laws. Let $\square^n$ be the standard n-cube.

Furthermore, we often want to consider precubical sets with designated start and end points. That is bipointed precubical sets, notated $\square_*^*$. That is, sets with a start and end.

## 2 Stardard Build

There are three essential structures on $\square_*^*$. The first is the wedge product, $\vee$. Given two $\square_*^*, P, Q$, we define $P, Q$ as the pushout Wedging cubes together turns out to be very useful. We define the serial wedge $\bigvee[a_1, ..., a_n]$ as the wedge of the $a_i$ cubes. Then, the second critical construction is, $\mathrm{Ch}(K)$, the category of serialWedge maps into $K$, with morphisms as $\square$ maps from $\bigvee a \longrightarrow \bigvee b$ that commute in a triangle with $K$. And then a third construction, $\mathrm{Refine}(K)$. A chain in $K$ is a finite sequence of cells of K, $[c_1, ..., c_n]$ such that $c_i$'s endpoint is $c_{i+1}$'s initial point, and $c_1$'s start agrees with $K$'s start, and $c_n$'s end agrees with $K$'s end. A refinement is a list substitution so that each

Futhermore, there are a few very helpful properties: Admits Altitude, and Non Self Linked. Look at the Configuration Spaces paper for details

A few things to note

**Lemma 2.1.** *The $\vee$ operator is a monoidal product on $\square_*^*$.*

**Lemma 2.2.** *The* Ch *construction is a lax monoidal functor on $\square_*^* \Rightarrow Cat$ with Cat's cartesian product.*

**Lemma 2.3.** *The categories* $\mathrm{Ch}(K)$ *and* $\mathrm{Refine}(K)$ *are equivalent as categories when $K$ admits altitudes and is NSL In this setting* Ch *is a strict monoidal functor.*

**Lemma 2.4.** *Serial wedges admit altitudes, and are NSL*

This is all the foundational material we need

## 3 Wedge Machinery

We will end up several times in a situtation where we know what to do on cubes, but want to work with serial wedges. In an effort to keep our proofs better factored, we build some machinery here will enables a more elegant lifting stategy.

**Lemma 3.1.** *Let $P$ be a functor* Cube $\Rightarrow Cat$*. Then there is an extension*

$$W(P) : (\square_*^*, \vee, 0) \Rightarrow (Cat, \oplus, \emptyset)$$

*which is a lax monoid functor extending $P$ via coyoneda. Moreover, if $P\square^0$ is empty, then for any serial wedge $\bigvee a$ we get an isomorphism*

$$W(P)(\bigvee a) \simeq \Sigma_i P(\square^{a_i})$$

*Proof.* The extension is, essentially, the Coend on $\square$. The isomorphism goes by induction. At each point, since $P(\square^0)$ is empty, apply $W(P)$ to the pushout diagram trivializes the shared point, so we end up with exactly the diagram of a coproduct. $\square$

**Lemma 3.2.** *Let $P$ be a functor* $\mathrm{Cube}^{op} \Rightarrow Cat$*, such that $P\square^0$ is a singleton. Then there is a*

$$W(P) : (\square_*^*, \vee, 0) \Rightarrow (Cat, \times, 1)$$

*is a lax monoidal functor extending $P$ via yoneda. Moreover, if $P(\square^0)$ is a singleton, then for any serial wedge $\bigvee a$, we get an isomorphism*

$$W(P)(\bigvee a) \simeq \Pi_i P(\square^{a_i})$$

*Proof.* The dual construction. Observe this time that if $P(\square^0)$ is unique, the coherence requirements for a wedge are trivial. $\square$

Together, these give us tools for lifting and decomposing cube maps to tensor maps.

So let us define the cases we care about

**Definition 3.3.** Let Coord : Cube $\Rightarrow$ $Cat$ send $\square^n \mapsto [n]$, and a map $f : \square^n \to \square^m$ is a face inclusion which induces a $[n] \to [m]$. This is a functor. And $[0]$ is empty, so it satisfies the lifting condition. This gives us $coordMap : (\bigvee a \longrightarrow \bigvee b) \to (\Sigma_i[a_i] \to \Sigma_j[b_j])$ directly as a monoidal functor. This tells us how coordinate positions in $\bigvee a$ translate to positions in $\bigvee b$.

**Lemma 3.4.** *For any $f : \bigvee a \longrightarrow \bigvee b$, the function $coordMap(f)$ is bijective.*

*Proof.* The proof needs first breaks up easily via $\vee$, so it suffices to show it for a map $\bigvee a \longrightarrow \square^b$. Then we prove this is an injection by an induction over the wedge, observing that in the cube, if a 0-cell $x$ can reach a 0-cell $y$, it always has more 1s in its coordinates. $\square$

Critically, this means that every coordinate in $\square^n$ only flips from 0 to 1 once for each $\bigvee a \longrightarrow \square^n$ as witnessed by $coordMap^{-1}$. This is the key property that will make our braids well-defined.

# 4 The Complexification Construction

The construction starts by observing that one dimensional serial wedges behave like topes, or chambers, from the hyperplane configuration world. That is, they are the maximally refined elements. Let's start by reviewing the story on the braid arrangement side.

Following Paris, but specifically for the braid arrangement. The $n-1$ braid arrangement is a family of hyperplanes in $\mathbb{R}^n$ of $\{x_i = x_j\}$ for $0 <= i < j <= n$. The idea is to give each plane an orientation, so that we can break up $\mathbb{R}^n$ into faces represented as functions **Face** $:= [n] \to \{-1, 0, 1\}$, the covectors. We can order covectors

$$x \leq y := \forall i, x(i) = 0 \vee x(i) = y(i)$$

which gives an order on **Face**. Maximal elements of this order are covectors that are never 0, which are precisely the complement of the hyperplanes. We call these maximal elements **topes**.

The main issue with **Face** is that it is both interesting and contractible. The goal of Salvetti's theorem is to build a structure on top of the same covectors that captures essentially the same information, but with a richer structure. Repeat the process, but with $\mathbb{C}$ instead of $\mathbb{R}$. We end up with the Salvetti Poset Sal.

$$\mathrm{Sal} = \{(F, C) \in \mathbf{Face} \times \mathbf{Face} \mid C \text{ is a tope} \wedge F \leq C\}$$

We need one more definition: the composition of two faces

$$F \circ G := i \mapsto \text{ if F(i) = 0 then } Y(i) \text{ else } X(i)$$

For a tope $C$ and face $F$, we define $C_F$ as the

$$(F, C) \preceq (F', C') := F \leq F' \wedge C' = F' \circ C$$

This poset induces a complex, and Salvetti's theorem tells us that it is isomorphic to the configuration space of $n$ points. So, its fundamental group is pure braids. Intuitively, each face get annotated with a direction, indicated by the tope. So when we paths cross through faces, they can cause orientation changes. These changes are what the braid words represent. And in fact, this direction information is what the Configuration Spaces paper retains as well, essentially breaking every tie at a face in every possible way.

This construction abstracts a bit, in a way that will make it convienent to construct for us. In particular, there's a functor $Tope : Face \Rightarrow \{X \in Face, X \text{ is a tope}\}$ where each face maps to its neighboring topes. And functions between faces induce orientation-preserving maps of topes. Then the salvetti poset is $\int Tope$.

To rebuild this generically on the precubical side, let's start with the first positive indicator.

**Definition 4.1.** $\mathrm{Ch}(\square^n) \simeq \mathbf{Face}$

**Definition 4.2.** A serial wedge is **one dimensional** if all its entries are 1

**Definition 4.3.** Let $\mathrm{Run}(K)$ be the full subcategory of $\mathrm{Ch}(K)$ such that the domains are one dimensional serial wedges.

**Lemma 4.4.** *Being one dimensional is preserved under $\vee$, so $\mathrm{Run}(K)$ is a monoidal functor. And in paritcular $List\mathbb{N} \Rightarrow \mathrm{Run}(\bigvee \cdot)$ is a lax monoidal functor.*

Now, the morphisms we want to build are a bit unusual, because they are built from degenerecies, which are not morphisms in Precubical. That is, we want to build a map from an op arrow in $\mathrm{Ch}(K)$ to runs that respect its's coordinate wise projections.

$$(\bigvee a \longrightarrow \bigvee b) \to (\mathrm{Run}(\bigvee b) \to \mathrm{Run}(\bigvee a))$$

Doing this directly is rather painful, so we build a bit a machinery to lift a cube-level operations to wedge level operations.

So specifically, let $\mathrm{Edge}(K)$ be the chains in $\mathrm{Refine}(K)$ with each cell 1 dimensional. Then $\mathrm{Run}(K) = \mathrm{Edge}(K)$.

Now, given a face embedding $(f : \square^n \longrightarrow \square^m)$, it defines a unique projection of cells of $m$ to cells of $n$, non-increasing on dimension. Call it $\hat{f}$. So we define

$$\rho : (f : \square^n \longrightarrow \square^m) \to (\mathrm{Run}[\square^m] \to (\mathrm{Run}[\square^n]))$$

via

$$\rho' : (f : \square^n \longrightarrow \square^m) \to (\mathrm{Refine}_1[\square^m] \to (\mathrm{Refine}_1[\square^n]))$$

by

$$[c_1, ..., c_k] \mapsto [\hat{f}(c_1), ..., \hat{f}(c_k)]$$

Note that this is indeed a one dimensional chain in $\square^n$ because $\hat{f}$ never increases dimension, and every projection respects initial/final points.

Now, $\rho$ is a nice. In particular,

**Lemma 4.5.** *$\rho : \mathrm{Cube}^{op} \Rightarrow Set$ is a functor, and $\rho(\square^0)$ is a singleton*

*Proof.* The proof of functorality mostly just exploits the fact that cubes are NSL and admit altitudes. So $\text{Refine}_1(\square^n) \simeq \text{Run}(\square^n)$ as sets. First, $\rho$ preserves identities, as the projection for with the identity embedding is also the identity. Second, if $f : \square^n \to \square^m$ and $g : \square^m \to \square^n$ are face embeddings, then their corresponding projections respect composition.

For the singleton, observe that there $\rho(\square^0) = \text{Run}[\square^0]$, but the only one dimensional serial wedge which maps into $\square^0$ is the the empty one. So indeed $\rho(\square^0)$ is a singleton. ☐

Now, the trick here is to observe that this makes $\rho$ a precubical set! Even better, it has a unique 0-cell, which brings us to the following results

**Lemma 4.6.** *If $P$ is $\square$, and has a unique 1-cell, then*

$$(\bigvee a) \longrightarrow P \simeq \Pi_i P[a_i]$$

*Proof.* The serial wedge is an iterated pushout over a point. But if the point is unique, then the pushout degenerates by its universal property into just one map per edge. Which is exactly a choice of appropriatly sized elements in P for each element of the wedge ☐

Futhermore, since Run is monoidal,

$$(\bigvee a) \longrightarrow \rho \simeq \Pi_i \text{Run}[a_i] \simeq \text{Run}[\bigvee a]$$

Formally, this make $\rho$ the classifying object for Run of wedges. Now our goal function

$$\theta : (\bigvee a \longrightarrow \bigvee b) \to \text{Run}(\bigvee b) \to \text{Run}(\bigvee a)$$

is equivalent to

$$(\bigvee a \longrightarrow \bigvee b) \to (\bigvee b \to \rho) \to (\bigvee a \to \rho)$$

which is just a composition.

now, that lets us define our desired functor

$$\text{Lines} : \text{Ch}(K)^{op} \Rightarrow \mathit{Set}$$

with $\text{Lines}(\bigvee a \longrightarrow K) \mapsto \text{Run}(a)$ on objects, and $\text{Lines}(f : \bigvee a \longrightarrow \bigvee b) \mapsto \theta(f)$ on morphisms. Lines's functoriality is inherited from $\theta$. Which finally brings us to

$$\text{Ch}^*(K) := \int \text{Lines}(K)$$

the category of elements for Lines.

## 5 Relationship to the Salvetti Complex

The key fact here is that our story mirrors the salvetti story, on the nose, for cubes. That is, the face complex is exactly $\text{Ch}(\square^n)$, and the salvetti complex is exactly $\text{Ch}^*(\square^n)$. I'd argue this explains why the braid invariant from 'Configuration Spaces' collapsed, and why this complexification survives. However, it does suggest that

1. The salvetti complex is also $\text{Ch}(Z)$, so the quotiont by the free action some is also related to the complexification story
2. $\text{Ch}^*(Z)$ is bigger than Braid, so something interesting is happening

## 6 The Braid Functor